

A Calibrated Selective Certificate for a Wall-Constrained Perception Gate

Anonymous

Abstract

Concept-guided robot perception can use an estimated room to restrict later door detection, but a hard geometric threshold can turn a small wall error into a changed decision. We isolate the first wall-distance gate of the concept-guided exploration system of Zapata et al. and connect it to the interface and calibration requirements identified by Zhao et al. For a frozen room fitter, exchangeable ground-truthed calibration episodes give a split-conformal upper radius for Hausdorff wall-set error. On the resulting room-level coverage event, the distance-to-set inequality certifies, simultaneously, every first-gate decision outside an explicit ambiguity band. The probability statement is marginal over rooms, applies only to the first gate, and does not certify later door detection, fallback accuracy, or robot motion. No episode-level calibration data are available in the source papers, so this is a theorem and validation protocol rather than a demonstrated calibrated robot system.

1 Introduction

Learning-augmented algorithms distinguish performance with useful and arbitrary predictions and require explicit accounting when guarantees pass between components [2]. Concept-guided exploration supplies a concrete, discontinuous interface: an estimated room wall set constrains a later door detector [1]. The detector retains a scan point p when

$$\text{dist}(p, W) \leq \delta, \quad \delta = 0.15 \text{ m}, \quad (1)$$

before constructing a polar scan, differentiating it, and pairing peaks. Zapata et al. report that noise can lead to accepted incorrect room models, while revision of accepted models is left for future work.

This paper asks a deliberately narrower question: can the first gate in (1) abstain selectively with a finite-sample statistical certificate? Our new result is a reduction from marginal, room-level Hausdorff coverage to simultaneous correctness of every non-abstained first-gate decision in a new exchangeable room. The reduction combines a standard split-conformal rank argument with the distance-to-set inequality; neither source paper states this problem-specific

chain. The claim is weaker than full selective safety for door perception: later processing couples scan points and remains uncertified.

The paper keeps four categories separate. Section 2 records what the two source papers establish. Section 3 gives the new result. Section 5 offers interpretation and a validation protocol, not further theorems. Sections 6 and 7 state limitations and open questions.

2 Results supplied by Q and P

Zhao et al. survey prediction cost, consistency and robustness, and the conditions needed to compose guarantees across interfaces [2]. In particular, their composition discussion requires a sensitivity or perturbation relation at an interface, and their calibration discussion warns that exchangeable split-conformal coverage is marginal and does not by itself control downstream cost. These statements motivate the proof obligation here, but Q does not give an accept, defer, or revision theorem for P’s robot system.

Zapata et al. implement asynchronous room and door concept agents that build a persistent scene graph, with estimated room walls constraining door detection [1]. Their first door-processing step is exactly (1). Filtering occurs before polar resampling, differentiation, thresholding, and peak pairing, so downstream candidates are not independent pointwise decisions. P supplies neither a Hausdorff-error coverage radius for its estimated walls nor episode-level wall estimates paired with reference walls. Its pose-graph covariance is not, without an additional argument, such a radius.

3 Calibrated selective gate

Let W^* and $\widehat{W}$ be non-empty wall sets in the same metric space, using the same convention for wall centreline or surface, spatial extent, and reference frame. Write

$$d_H(\widehat{W}, W^*)$$

for their Hausdorff distance. For every point p ,

$$\left| \text{dist}(p, \widehat{W}) - \text{dist}(p, W^*) \right| \leq d_H(\widehat{W}, W^*). \quad (2)$$

Proposition 1 (Deterministic margin transfer). *If $d_H(\widehat{W}, W^*) \leq \varepsilon$, then*

$$\text{dist}(p, \widehat{W}) + \varepsilon \leq \delta \implies \text{dist}(p, W^*) \leq \delta, \quad (3)$$

$$\text{dist}(p, \widehat{W}) - \varepsilon > \delta \implies \text{dist}(p, W^*) > \delta. \quad (4)$$

Points satisfying neither antecedent are called ambiguous.

Proof. Both implications follow directly from (2). The weak inequality in (3) and strict inequality in (4) match P’s acceptance convention in (1). $\square$

Without a positive margin from the threshold, gate membership is unstable under arbitrarily small wall displacement. For any positive displacement, a point can be placed sufficiently close to the threshold so that translating a wall by no more than that displacement flips the binary gate value. Thus the binary gate is not globally Lipschitz as a map from Hausdorff wall error to membership error.

The deterministic statement is not operational until ε is obtained. Freeze the complete room fitter independently of calibration. For n exchangeable, ground-truthed calibration episodes, define

$$R_i = d_H(\widehat{W}_i, W_i^*), \quad k = \lceil (n+1)(1-\alpha) \rceil. \quad (5)$$

Let ε_α be the k th order statistic of $R_1, \dots, R_n$, with $\varepsilon_\alpha = +\infty$ if $k > n$.

Theorem 1 (Marginal simultaneous first-gate certificate). *Suppose the frozen fitter is applied to calibration episodes and one new episode such that their scores are exchangeable. Then*

$$\Pr\{d_H(\widehat{W}_{\text{new}}, W_{\text{new}}^*) \leq \varepsilon_\alpha\} \geq 1 - \alpha. \quad (6)$$

On the event inside (6), every new-room point certified by (3) or (4), using ε_α , receives the same first-gate decision as it would under W_{new}^ . Consequently,*

$$\Pr\left\{\begin{array}{c} \text{all non-ambiguous first-gate decisions agree} \\ \text{with the reference-wall gate} \end{array}\right\} \geq 1 - \alpha. \quad (7)$$

Proof. Exchangeability makes the new score’s rank among the $n+1$ scores uniform up to conservative handling of ties. The order-statistic choice in (5), including the infinite-radius convention, gives (6). On that single room-level event, (2) holds for every scan point. The deterministic margin transfer then proves all certified decisions simultaneously. No union bound over scan points is used. $\square$

Equation (7) is marginal over exchangeable episodes. It is not conditional coverage for a given room, geometry class, or noise regime.

4 Related work and novelty status

Q discusses split conformal as a possible distribution-free wrapper under exchangeability, while stressing that marginal coverage alone does not imply a downstream guarantee [2]. P studies concept-guided room and door perception but does not calibrate wall-set Hausdorff error or prove a selective gate certificate [1]. Our specific contribution is the intervening geometric reduction from one room-level coverage event to simultaneous agreement of non-abstained first-gate decisions.

The targeted search record is reported in `search.md`. Network name resolution was unavailable in this round, so arXiv and DOI landing pages for additional candidates could not be read or verified. The paper therefore makes no

claim that the reduction is absent from the external literature. This limitation changes the novelty status: the contribution is established relative to Q and P and the supplied peer record, not as a priority claim over all prior work.

5 Interpretation and validation protocol

The theorem suggests an interface contract rather than an end-to-end robot guarantee. Outside the ambiguity band, estimated-wall processing agrees with the reference-wall first gate on the covered room event. Inside the band, a system must abstain or invoke a separately justified procedure. This is an interpretation of the theorem, not evidence that any particular fallback is accurate or efficient.

Pointwise fallback is not supported. Removing a point changes the polar scan’s adjacency and can alter derivatives and peak pairs, so a fallback must cover a whole scan or a declared dependency region with fixed grid, interpolation, and boundary rules. No such regional contract is implemented or evaluated in the supplied material.

The smallest informative experiment is to predeclare a deployment population and wall-set convention, freeze P’s fitter, and collect disjoint ground-truthed calibration and test rooms in its simulator. It should report test-room coverage, $\varepsilon_\alpha/\delta$, and the fraction of declared dependency regions sent to fallback, alongside compute and motion as separate quantities, door recall, and topology error. An outcome with $\varepsilon_\alpha \geq 0.15$ m or near-total regional fallback is a valid but operationally negative result.

6 Limitations

The theorem has the following boundaries.

- Exchangeability is essential. Distribution shift, a new building or sensor regime, and dependent adaptive sampling are not covered.
- Coverage is marginal over rooms, not conditional safety for an individual room or rare layout.
- Estimated and reference walls must share a wall-set convention, extent, and frame. Otherwise the calibration score is not the claimed quantity.
- Only the first distance gate is certified. Derivative, peak-pair, width, segment, final door, and topology decisions are not certified.
- The result does not establish fallback correctness or cost, and it does not establish a competitive guarantee for policy-dependent robot motion.
- The source material contains no calibration episodes. Thus the radius, achieved coverage, and abstention cost cannot be computed here.

- The external prior-work search could not be completed because scholarly landing pages were unreachable, so literature-level novelty remains unresolved.

These limitations prevent the phrases “safe door detector” and “calibrated robot system” from being justified by the present result.

7 Open questions

The first open question is whether a defensible deployment population and independent episode unit can be specified for P’s active exploration process. The second is empirical non-vacuity: Q and P do not show whether the calibrated radius is below the gate threshold or whether dependency expansion forces nearly total fallback. The third is how to define and validate a regional fallback whose own accuracy and work are independently characterised. The fourth is whether later discontinuities admit compatible margin conditions without strengthening the present first-gate theorem. Finally, a complete external literature search is needed to determine whether the geometric coverage-to-gate reduction or its main proof step has already been published.

8 Conclusion

The hard wall gate in P does not automatically preserve small geometric errors, but it admits a precise selective certificate. Under a frozen fitter, exchangeable calibration, and a fixed wall-set convention, a split-conformal Hausdorff radius gives marginal room-level coverage; on that event, the distance-to-set inequality transfers coverage simultaneously to every non-abstained first-gate decision. This is a first-interface theorem and validation protocol, not a claim about complete door perception or robot safety.

References

- [1] Noé Zapata, Gerardo Pérez, Alejandro Torrejón, Pedro Núñez, and Pablo Bustos. Concept-guided exploration: Building persistent, actionable scene graphs, 2026.
- [2] Hailiang Zhao, Xueyan Tang, Peng Chen, Zenong Ye, Tianjun Zhu, Jianwei Yin, and Shuiguang Deng. Learning-augmented algorithms: Guarantees, construction mechanisms, and system-level implications, 2026.